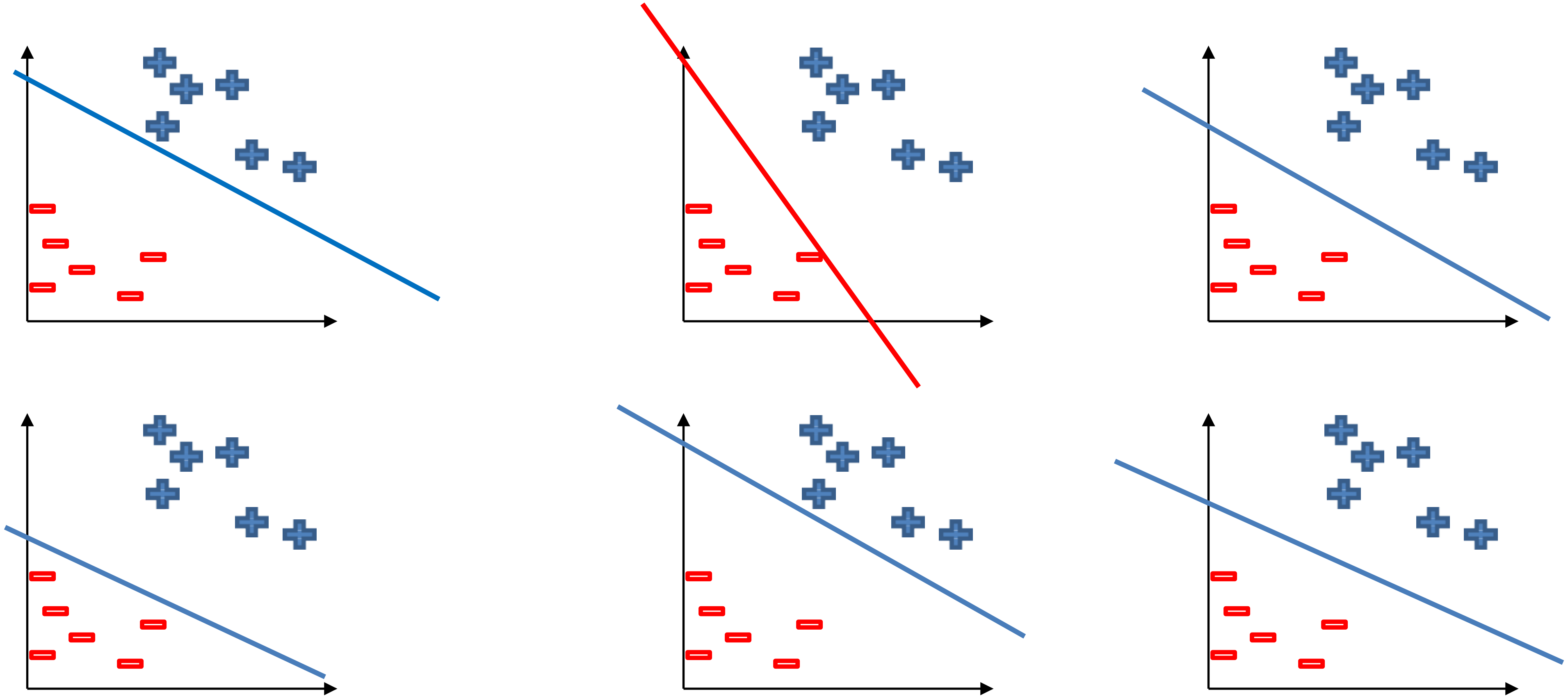
Support Vector Machine

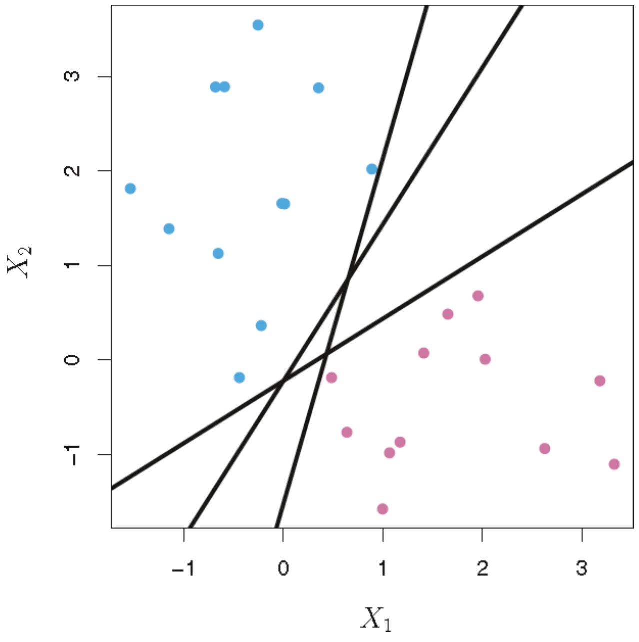
Lecture 20

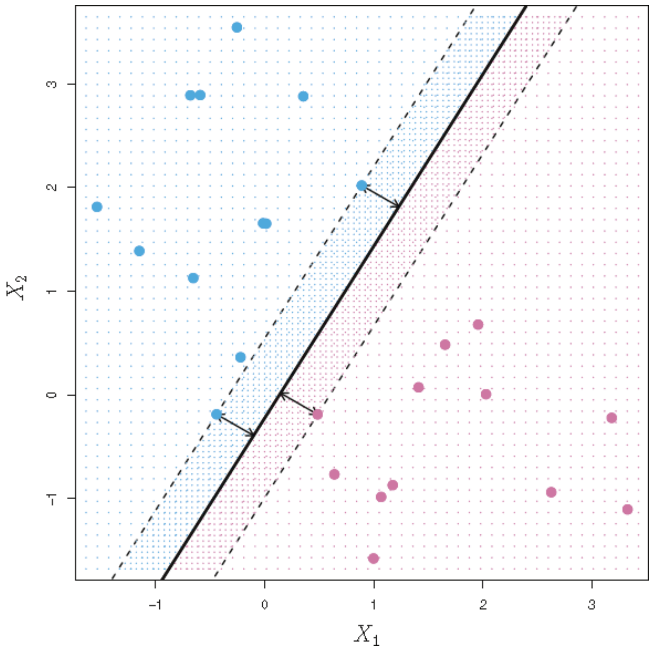
Perceptron



Support Vector Machine

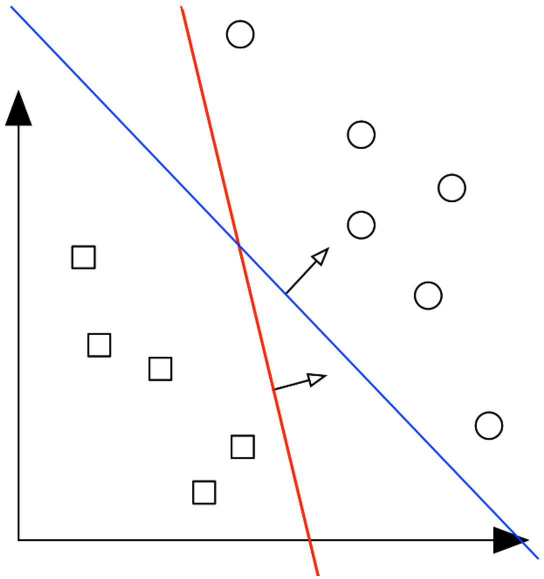
- The Support Vector Machine (SVM) is a linear classifier that can be viewed as an extension of the Perceptron developed by Rosenblatt in 1958.
  - The Perceptron guaranteed that you find a hyperplane if it exists.
  - The SVM finds the maximum margin separating hyperplane.

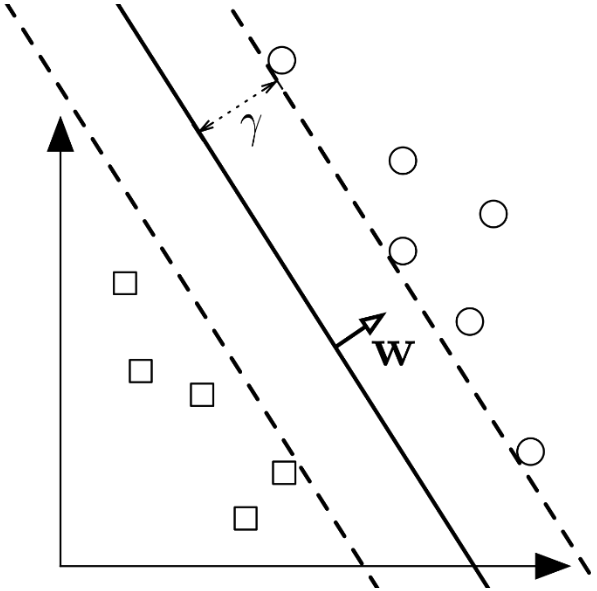




Support Vector Machine

- If a data set is linearly separable, typically there are infinitely many separating hyperplanes. What is the best separating hyperplane?
- **SVM Answer:** The hyperplane that maximizes the distance to the closest data points from both classes – the hyperplane with <u>maximum margin</u>





Support Vector Machines

- An optimization problem
- We maximize the margin: the distance separating the closest pair of data points belonging to opposite classes.
- These points are called the support vectors, because they are the data observations that “support”, or determine, the decision boundary.
  - These are also the samples that are most difficult to classify.
- To train an SVM classifier, we find the maximal margin hyperplane, or optimal separating hyperplane, which optimally separates the two classes in order to generalize to new data and make accurate classification predictions.

Innovation

Logistic Regression

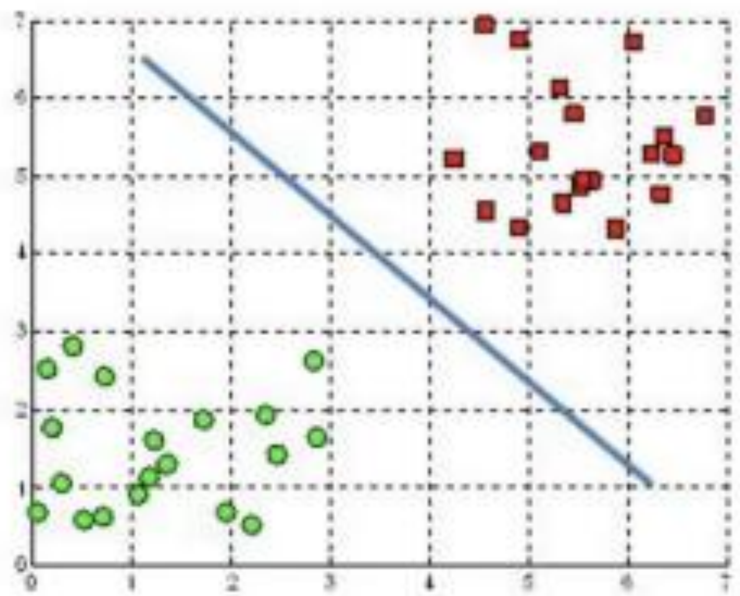
- Used for classification alone
- Probabilistic classifier that o/p probability estimates for k classes and assigns the label of the class with the highest probability
- Use statistical properties of data
- Identifies a single hyperplane by using the idea of perceptron logic
- Logistic Regression is a linear model and work on linear data only
- Sensitive to outliers and prone to overfitting

Support Vector Machine

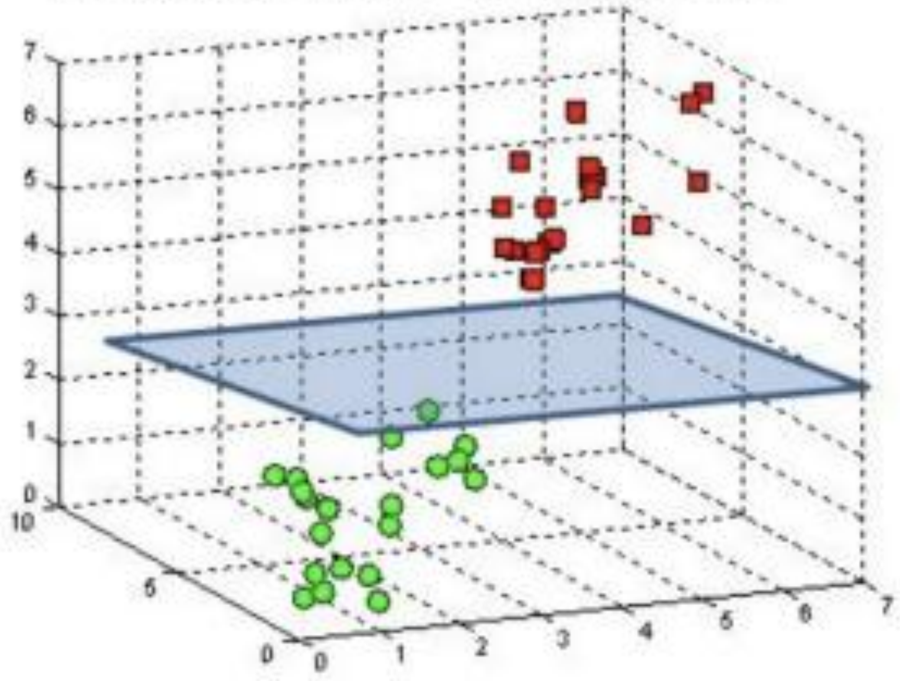
- Used for classification as well as regression
- Deterministic classifier that o/p a hard label without providing the probability estimate for k classes
- Use geometric properties of data and is also called maximum margin classifier
- Identifies three hyperplanes (extends the idea of perceptron logic)
- SVM is a linear model but can work on nonlinear data as well (kernel trick)
- Robust to outliers and risk of overfitting is relatively lower

Hyperplanes as Decision Boundaries

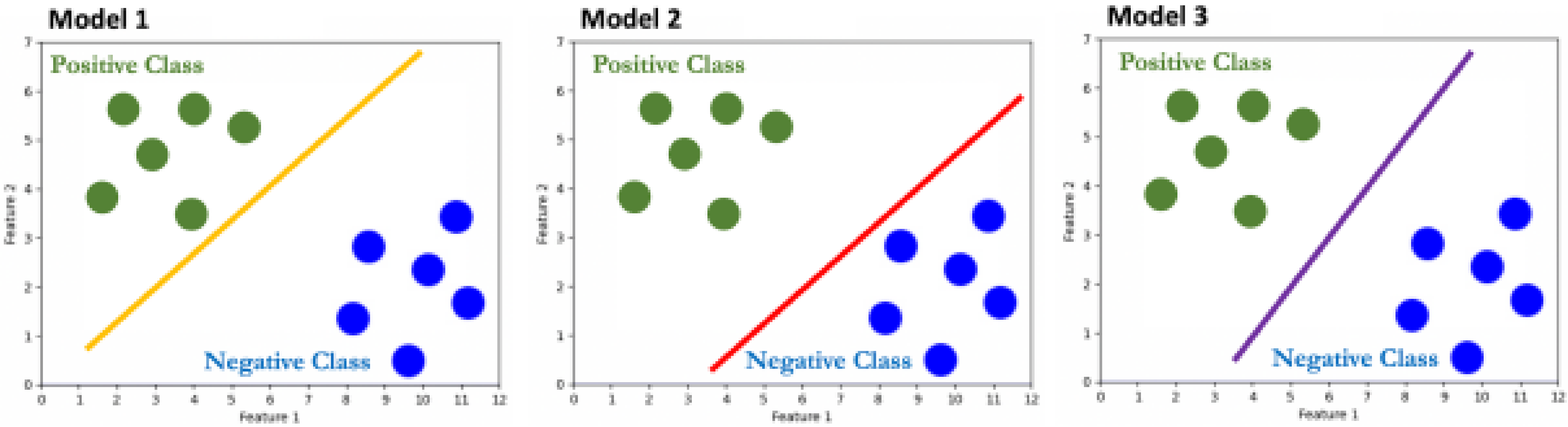
A hyperplane in $\mathbb{R}^2$ is a line



A hyperplane in $\mathbb{R}^3$ is a plane



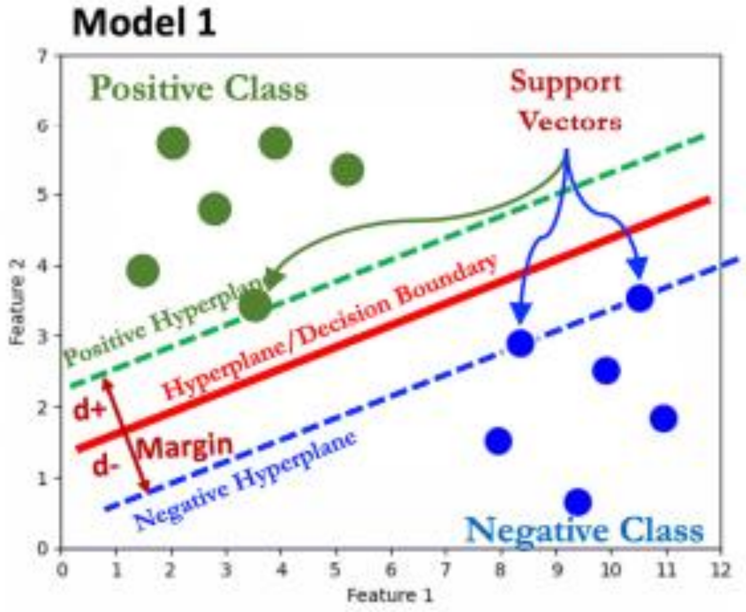
How to Select the Best Hyperplane?

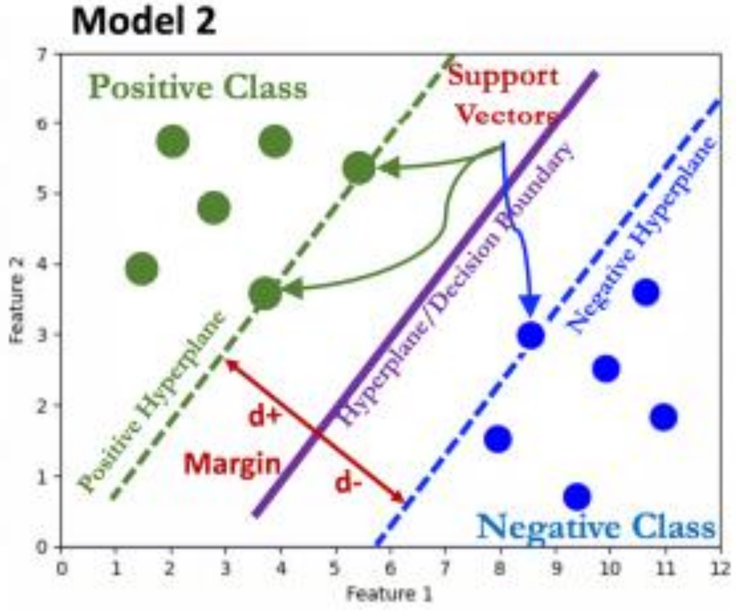


- If the hyperplane lies very close to either the positive or negative data points, a new test point may get mis-classified

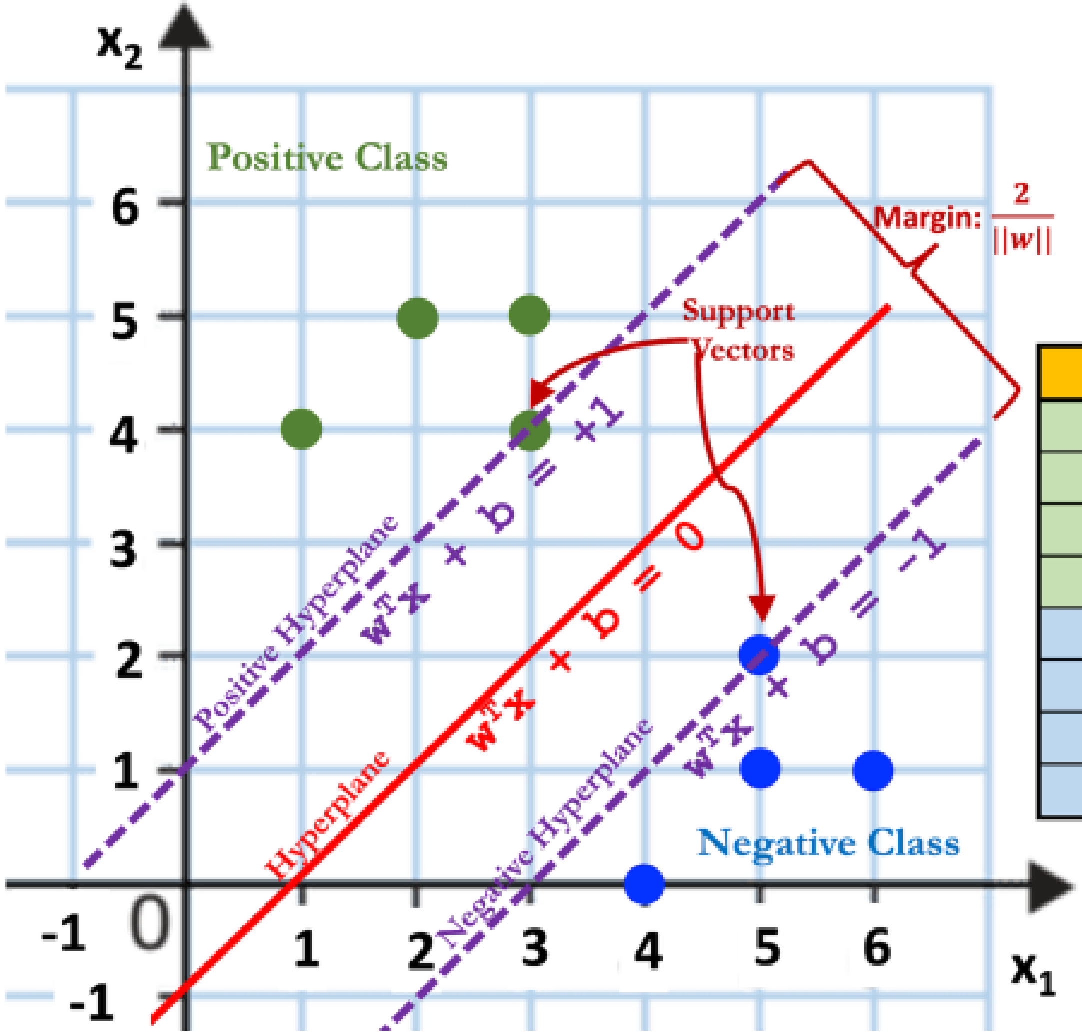
How does SVM Work (A Geometric Intuition)?

SVC is also called Maximum Margin Classifier as it tries to maximize the margin that separates the two classes



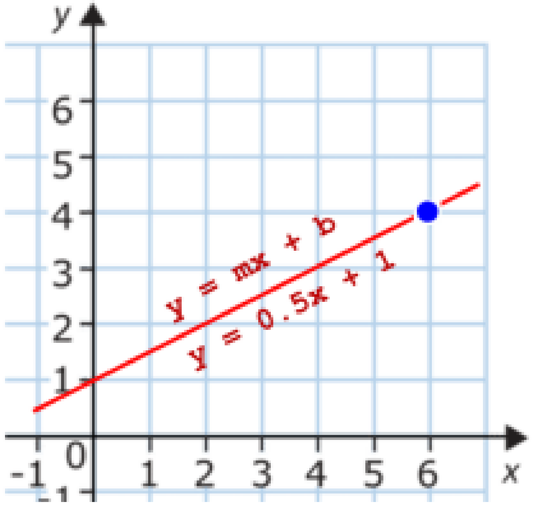


Support Vector Machines



x1	x2	Class
1	4	1
2	5	1
3	5	1
3	4	1
6	1	-1
4	0	-1
5	2	-1
5	1	-1

Math Behind SVM



$y = 0.5x + 1$

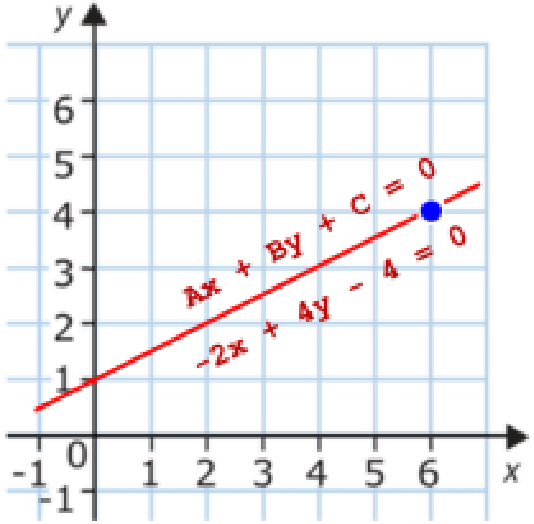
$-0.5x + y - 1 = 0$

$-2x + 4y - 4 = 0$

$A=-2,\ B=4,\ C=-4$

$y = -\frac{A}{B}x - \frac{C}{B}$

$y = -\frac{-2}{4}x - \frac{-4}{4}$



$$Ax_1 + Bx_2 + C = 0$$

$$w_1x_1 + w_2x_2 + w_0 = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$$

$$w_1x_1 + w_2x_2 + w_3x_3 + \cdots + w_mx_m + w_0 = 0$$

$$\vec{w}\vec{x} + w_0 = 0$$

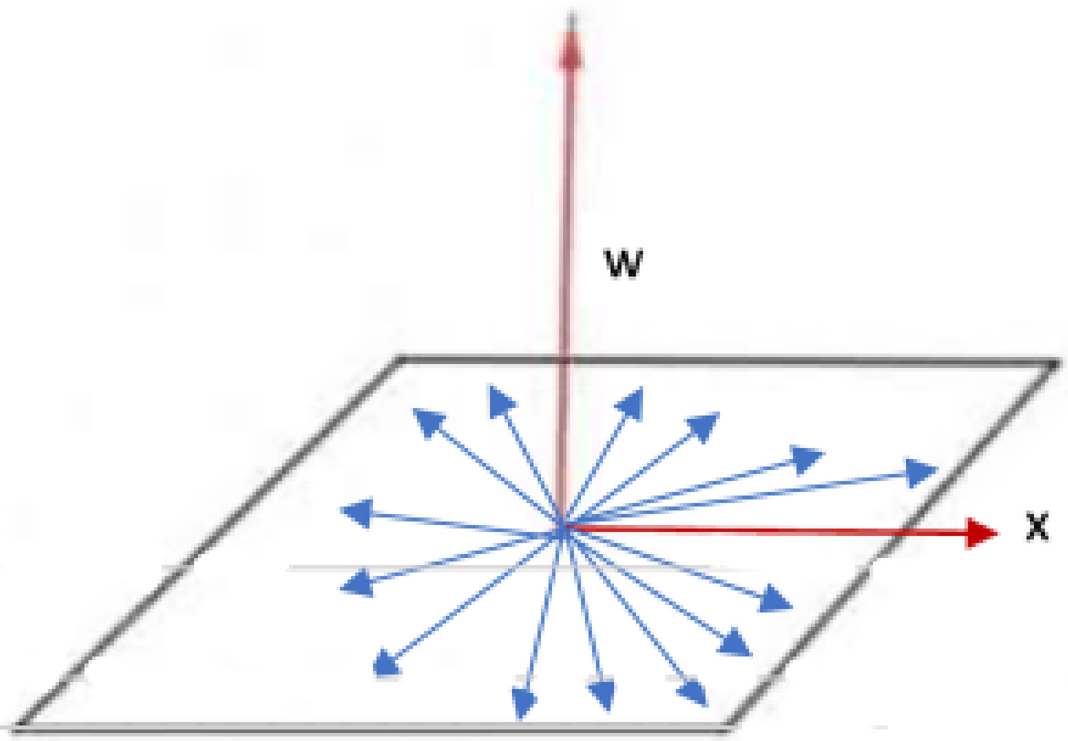
$$\vec{w}\vec{x} = 0$$

$$w^T x = 0$$

$$\begin{bmatrix} w_1 & w_2 & w_3 & \cdots & w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = 0$$

Interpretation of Equation of Vector

- Since the dot product of $\vec{w}$ and $\vec{x}$ is zero, that means the angle between $\vec{w}$ and $\vec{x}$ is 90 degrees, i.e., they are orthogonal vectors
- Therefore, $\vec{w}$ is a vector which will always be perpendicular to the hyperplane (decision boundry).

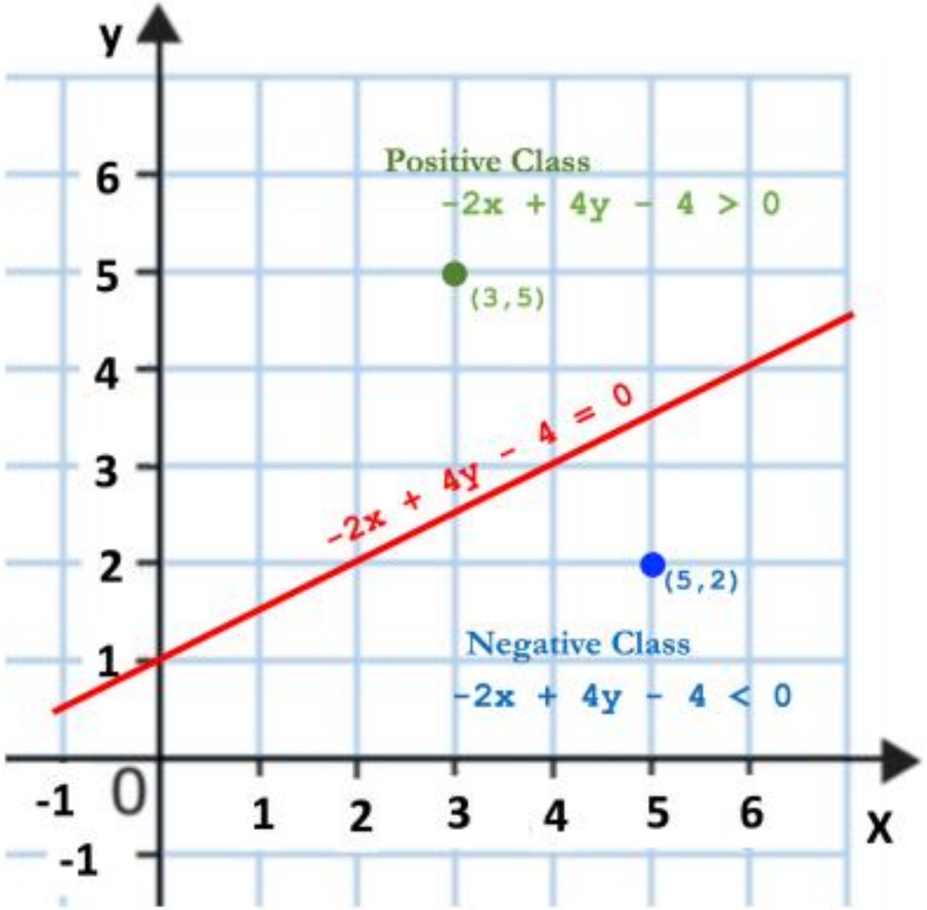


Classifying a New Data Point

- For any vector $\overrightarrow{x}$ in this entire vector space, it's corresponding predicted label $\hat{y}$ is $+1$ if $w^T x + b \geq 0$ and -1 if $w^T x + b < 0$

- This can formally be written as follows:

$$\hat{y} = \begin{cases} +1, & \text{if} \quad w^T x + b \geq 0 \\ -1, & \text{if} \quad w^T x + b < 0 \end{cases}$$



https://www.desmos.com/calculator

Constraint for Hard SVM

- We need to maximize the margin, such that
  - NO training point is below the Positive Hyperplane :

$$\vec{w}\,\vec{x} + b \geq 1$$

  - NO training point is be above the Negative Hyperplane :

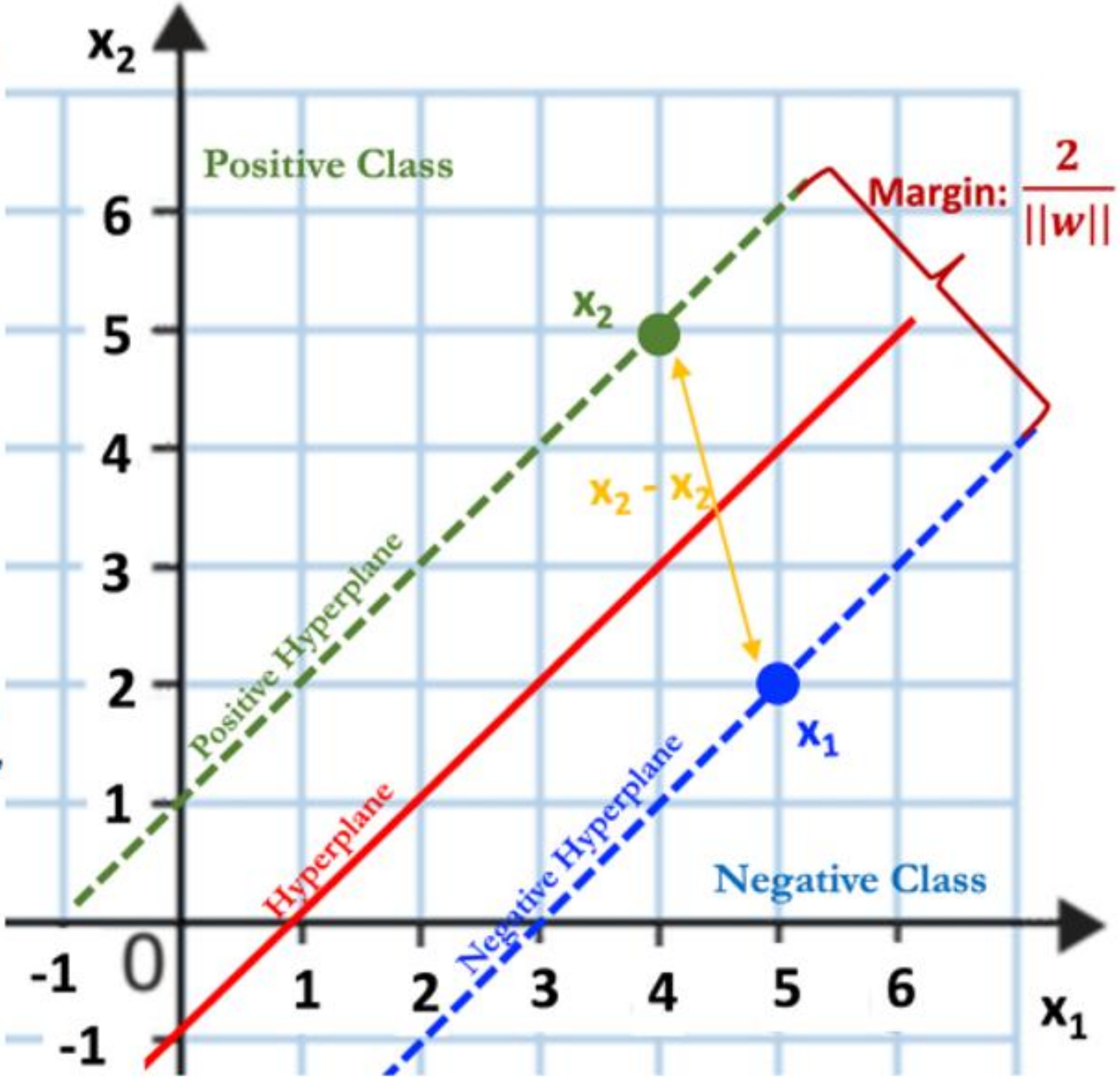
$$\vec{w}\,\vec{x} + b < -1$$

- Simplify the above two equation to one, assuming that the label of all the positiv training points is $+1$ and the label of all the negative training points is -1:

$$\hat{y}_i = y_i(\vec{w}\,\vec{x_i} + b) \geq 1$$

Formula to Calculate the Margin

- The shortest distance between x_1 and x_2 will be perpendicular to both the positive and negative hyperplanes.
- Now we do have a $\vec{w}$, which is perpendicular to both rather all the three hyperplanes, whose unit vector (vector having a magnitude of one) can be obtained by dividing $\vec{w}$ by its magnitude, i.e., $\|w\|$.



Formula to Calculate the Margin

$$d = (x_2 - x_1) . \frac{\vec{w}}{\|w\|}$$

$$d = \frac{x_2\vec{w} - x_1\vec{w}}{\|w\|}$$

$$d = \frac{\vec{w}x_2 - \vec{w}x_1}{\|w\|} \qquad \cdots \qquad (i)$$

- We already know that for all support vectors $y_i(\vec{w}\vec{x_i} + b) = 1$ holds.
  - Since x_2 is a positive support vector having y_i of $+1$, so $\vec{w}\vec{x_2} = 1 - b$
  - Since x_1 is a negative support vector having y_i of -1, so $\vec{w}\vec{x_1} = -1 - b$

- Substituting these values in equation (i):

$$d = \frac{1 - b - (-1 - b)}{\|w\|}$$

$$d = \frac{2}{\|w\|} \qquad \cdots \qquad (ii)$$

Mathematical Formulation of SVM

$$argmax_{(w^*,b^*)} \ \frac{2}{||w||}, \text{such that} \quad y_i(\vec{w}\vec{x_i} + b)$$

- Since 2 is a constant, so we can ignore it and the above expression can be writen as:

$$argmax_{(w^*,b^*)} \ \frac{1}{||w||}, \text{such that} \quad y_i(\vec{w}\vec{x_i} + b$$

- To change above into a minimization problem, we can write as:

$$argmin_{(w^*,b^*)} \ ||w||, \text{such that} \quad y_i(\vec{w}\vec{x_i} + b)$$

- For further ease of calculation, we can write above as:

$$argmin_{(w^*,b^*)} \ \frac{1}{2}||w||^2, \text{such that} \quad y_i(\vec{w}\vec{x_i} +$$

Mathematical Formulation of SVM

$$argmax_{(w^*,b^*)} \ \frac{2}{||w||}, \text{ such that } \quad y_i(\vec{w}\vec{x_i} + b)$$

- Since 2 is a constant, so we can ignore it and the above expression can be writen as:

100$ Question:

$$argmax_{(w^*,b^*)} \ \frac{1}{||w||}, \text{ such that } \quad y_i(\vec{w}\vec{x_i} + b$$

How to find the values of $\vec{w}$ and b for the best decision boundry that maximizes the distance between the positive and negative hyperplanes?

- To change above in

$argmin_{(w}$

One of the way to solve the above constraint optimisation problem is to add one more variable (called Lagrange multiplier) for each constraint and solve it.

- For further ease

$argmin_{(w}$ 2

Thank You..!